

PRACTICE SHEET 1

1. If 80% of A = 50% of B and B = x% of A, then the value of x is :
 (1) 400 (2) 300
 (3) 160 (4) 150
2. If x is 80% of y, what percent of x is y ?
 (1) 75% (2) 80%
 (3) 100% (4) 125%
3. If 8% of x is the same as 4% of y, then 20% of x is the same as:
 (1) 10% of y (2) 16% of y
 (3) 80% of y (4) 50% of y
4. A student multiplied a number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the percentage error in the calculation?
 (1) 44% (2) 34%
 (3) 54% (4) 64%
5. If p% of p is 36, then p is equal to :
 (1) 3600 (2) 600
 (3) 60 (4) 15
6. 2 is what percent of 50?
 (1) 2 (2) 2.5
 (3) 4 (4) 5
7. $\frac{2}{3}$ is what percent of $\frac{1}{3}$?
 (1) 50% (2) $33\frac{1}{3}\%$
 (3) 150% (4) 200%
8. 0.15% of $33\frac{1}{3}\%$ of 10000 is :
 (1) 5 (2) 150
 (3) 0.05 (4) 105
9. 30% of x is 72. The value of x is:
 (1) 216 (2) 240
 (3) 480 (4) 640
10. If 15% of (A + B) = 25% of (A - B), then what per cent of B is equal to A?
 (1) 10% (2) 60%
 (3) 200% (4) 400%
11. A supply of juice lasts for 35 days. If its use is increased by 40% the number of days would the same amount of juice lasts, is
 (1) 25 days (2) 30 days
 (3) 24 days (4) 27 days
12. If 60% of A = 30% of B, B = 40% of C, C = x% of A, then value of x is
 (1) 200 (2) 500
 (3) 800 (4) 300
13. In an office, 40% of the staff is female. 70% of the female staff and 50% of the male staff are married. The percentage of the unmarried staff in the office is
 (1) 64 (2) 60
 (3) 54 (4) 42
14. 50% of a number when added to 50 is equal to the number. The number is
 (1) 50 (2) 75
 (3) 100 (4) 150
15. $83\frac{1}{3}\%$ of Rs. 90 is equal to 60% of ?
 (1) Rs. 123 (2) Rs. 124
 (3) Rs. 122 (4) Rs. 125
16. 51% of a whole number is 714. 25% of that number is
 (1) 350 (2) 450
 (3) 550 (4) 250
17. What is to be added to 15% of 160 so that the sum may be equal to 25% of 240
 (1) 24 (2) 84
 (3) 60 (4) 36
18. What per cent of a day is 30 minutes?
 (1) 2.83 (2) 2.083
 (3) 2.09 (4) 2.075
19. A basket contains 300 mangoes. 75 mangoes were distributed among some students. Find the percentage of mangoes left in the basket.
 (1) 70% (2) 72%
 (3) 76% (4) 75%
20. The weights of two iron balls are 3.5 kg and 7.5 kg. What is the percentage weight of the first ball with respect to second ball?
 (1) $46\frac{2}{3}\%$ (2) 35%

(3) $46\frac{1}{3}\%$

(4) 45%

21. If x is less than y by 25% then y exceed x by:

(1) $33\frac{1}{3}\%$

(2) 25%

(3) 75%

(4) $66\frac{2}{3}\%$

22. If x is 10% more than y , then by what per cent is y less than x ?

(1) $9\frac{1}{11}\%$

(2) $7\frac{1}{11}\%$

(3) $8\frac{1}{11}\%$

(4) $10\frac{1}{11}\%$

23. If A's height is 10% more than B's height, by how much per cent less is B's height than that of A?

(1) 10%

(2) $10\frac{1}{9}\%$

(3) $10\frac{1}{11}\%$

(4) $9\frac{1}{11}\%$

24. B got 20% marks less than A. What per cent marks did A got more than B?

(1) 20%

(2) 25%

(3) 12%

(4) 80%

25. If x earns 25% more than y . What percent less does y earn than x ?

(1) 16%

(2) 10%

(3) 20%

(4) 25%

26. The difference of two numbers is 20% of the larger number. If the smaller number is 20, the larger number is :

(1) 25

(2) 45

(3) 50

(4) 80

27. If a number x is 10% less than another number y and y is 10% more than 125, then x is equal to :

(1) 150

(2) 143

(3) 140.55

(4) 123.75

28. Two numbers are respectively $12\frac{1}{2}\%$ and 25% more than a third number. The first number as percentage of second number is

(1) 50

(2) 60

(3) 75

(4) 90

29. Which number is 40% less than 90?

(1) 36

(2) 54

(3) 50

(4) 60

30. Two numbers are less than a third number by 30% and 37% respectively. The per cent by which the second number is less than the first is

(1) 10%

(2) 7%

(3) 4%

(4) 3%

ANSWER KEY

1.	(3)	2.	(4)	3.	(1)	4.	(4)	5.	(3)	6.	(3)	7.	(4)	8.	(1)	9.	(2)	10.	(4)
11.	(1)	12.	(2)	13.	(4)	14.	(3)	15.	(4)	16.	(1)	17.	(4)	18.	(2)	19.	(4)	20.	(1)
21.	(1)	22.	(1)	23.	(4)	24.	(2)	25.	(3)	26.	(1)	27.	(4)	28.	(4)	29.	(2)	30.	(1)

PRACTICE SHEET 2

1. If the total monthly income of 16 persons is 80,800 and the income of one of them is 120% of the average income, then his income is

(1) 5,050

(2) 6,060

(3) 6,160

(4) 6,600

2. Mita's income is 25% more than that of Sita. What percent is Sita's income less than that of Mita?

(1) 25%

(2) 24%

(3) $22\frac{1}{2}\%$

(4) 20%

3. A man spends $12\frac{1}{2}\%$ of his salary on items of daily use and 30% of the remainder on house rent. After that he is left with 2940. How much is his salary?

- (1) 4800 (2) 5200
(3) 4500 (4) 4000
4. The monthly income of a person was 13,500 and his monthly expenditure was 9,000. Next year his income increased by 14% and his expenditure increased by 7%. The per cent increase in his savings was
(1) 7% (2) 21%
(3) 28% (4) 35%
5. A worker suffers a 20% cut in his wages. He may regain his original wages by obtaining a rise of
(1) 27.5% (2) 25.0%
(3) 22.5% (4) 20.0%
6. Given that 10% of A's income = 15% of B's income = 20% of C's income. If sum of their income is 7800, then B's income is:
(1) 3600 (2) 3000
(3) 2400 (4) 1800
7. If A's income is 25% less than B's income, by how much percent is B's income more than that of A ?
(1) 25% (2) 30%
(3) $33\frac{1}{3}\%$ (4) $66\frac{2}{3}\%$
8. A's salary is 50% more than that of B. How much per cent is B's salary less than that of A?
(1) 50% (2) $33\frac{1}{3}\%$
(3) 45% (4) $66\frac{2}{3}\%$
9. Tulsiram's salary is 20% more than that of Kashyap. If Tulsiram saves 720 which is 4% of his salary, then Kashyap's salary is
(1) 15,000 (2) 12,000
(3) 10,000 (4) 22,000
10. A's salary is 40% of B's salary and B's salary is 25% of C's salary. What percentage of C's salary is A's salary?
(1) 5% (2) 10%
(3) 15% (4) 20%
11. The price of sugar is increased by 20%. If the expenditure on sugar has to be kept the same as earlier, the ratio between the reduction in consumption and the original consumption is
(1) 1 : 3 (2) 1 : 4
(3) 1 : 6 (4) 1 : 5
12. Rama's expenditures and savings are in the ratio 5 : 3. If her income increases by 12% and expenditure by 15%, then by how much per cent do her savings increase?
(1) 12% (2) 7%
(3) 8% (4) 13%
13. The ratio of two numbers is 4:5 when the first is increased by 20% and the second is decreased by 20%, the ratio of the resulting numbers is
(1) 4 : 5 (2) 5 : 4
(3) 5 : 6 (4) 6 : 5
14. If 60% of A = $\frac{3}{4}$ of B, then A : B is
(1) 9 : 20 (2) 20 : 9
(3) 4 : 5 (4) 5 : 4
15. If A exceeds B by 60% and B is less than C by 20%, then A : C is
(1) 32 : 25 (2) 25 : 32
(3) 8 : 5 (4) 4 : 5
16. If 30% of (B - A) = 18% of (B + A), then the ratio A : B is equal to
(1) 4 : 1 (2) 1 : 4
(3) 5 : 4 (4) 5 : 9
17. If 60% of A = 30% of B, B = 40% of C, C = x% of A, then value of x is
(1) 200 (2) 500
(3) 800 (4) 300
18. In an office, 40% of the staff is female. 70% of the female staff and 50% of the male staff are married. The percentage of the unmarried staff in the office is
(1) 64 (2) 60
(3) 54 (4) 42
19. 50% of a number when added to 50 is equal to the number. The number is
(1) 50 (2) 75
(3) 100 (4) 150
20. $83\frac{1}{3}\%$ of Rs. 90 is equal to 60% of ?
(1) Rs. 123 (2) Rs. 124
(3) Rs. 122 (4) Rs. 125

- 21.** 51% of a whole number is 714. 25% of that number is
 (1) 350 (2) 450
 (3) 550 (4) 250
- 22.** In a marriage party 32% are women, 54% are men and there are 196 children. How many men are there in the marriage party?
 (1) 756 (2) 448
 (3) 332 (4) 324
- 23.** What per cent of a day is 30 minutes?
 (1) 2.83 (2) 2.083
 (3) 2.09 (4) 2.075
- 24.** A basket contains 300 mangoes. 75 mangoes were distributed among some students. Find the percentage of mangoes left in the basket.
 (1) 70% (2) 72%
 (3) 76% (4) 75%
- 25.** The weights of two iron balls are 3.5 kg and 7.5 kg. What is the percentage weight of the first ball with respect to second ball?
 (1) $46\frac{2}{3}\%$ (2) 35%
 (3) $46\frac{1}{3}\%$ (4) 45%
- 26.** If A earns 25% more than B then how much percent does B earn less than A :
 (1) 15% (2) 20%
 (3) 25% (4) 30%
- 27.** What per cent of 1 day is 36 minutes?
 (1) 25% (2) 2.5%
 (3) 3.6% (4) 0.25%
- 28.** One number is 25% of another number. The larger number is 12 more than the smaller. The larger number is
 (1) 48 (2) 16
 (3) 4 (4) 12
- 29.** The number of students in a class is increased by 20% and the number now becomes 66. Initially the number was
 (1) 45 (2) 50
 (3) 55 (4) 60
- 30.** In a school there are 1500 students, 44% of them are girls. Monthly fee of each boy is Rs. 540 and the fee of each girl is 25% less than that of a boy. The sum of fees of boys and girls both is
 (1) Rs. 720600 (2) Rs. 720800
 (3) Rs. 720900 (4) Rs. 721000

ANSWER KEY

1.	(2)	2.	(4)	3.	(1)	4.	(3)	5.	(2)	6.	(3)	7.	(3)	8.	(2)	9.	(1)	10.	(2)
11.	(3)	12.	(2)	13.	(4)	14.	(4)	15.	(1)	16.	(2)	17.	(2)	18.	(4)	19.	(3)	20.	(4)
21.	(1)	22.	(1)	23.	(2)	24.	(4)	25.	(1)	26.	(2)	27.	(2)	28.	(2)	29.	(3)	30.	(3)

PRACTICE SHEET 3

- 1.** Salary of a person is first increased by 20%, then it is decreased by 20%. Percentage change in his salary is:
 (1) 4% decreased (2) 4% increased
 (3) 8% decreased (4) 20% increased
- 2.** A number is increased by 20% and then it is decreased by 10%. Find the net increase or decrease per cent.
 (1) 10% increase (2) 10% decrease
 (3) 8% increase (4) 8% decrease
- 3.** The tax imposed on an article is decreased by 10% and its consumption increases by 10%. Find the percentage change in revenue from it.
 (1) 10% increase (2) 2% decrease
 (3) 1% decrease (4) 11% increase
- 4.** The price of an article was decreased by 10% and again reduced by 10%. By what per cent should the price have been reduced once, in order to produce the same effect as these two successive reductions?
 (1) 15% (2) 19%
 (3) 20% (4) 25%

5. If price of a book is first decreased by 25% and then increased by 20%, the net change in the price of the book will be
 (1) 10% decrease (2) 5% decrease
 (3) no change (4) 5% increase
6. The price of an article is reduced by 25% but the daily sale of the article is increased by 30%. The net effect on the daily sale receipts is
 (1) 2.5% increase (2) 2.5% decrease
 (3) 2% increase (4) 2% decrease
7. Two successive price increases of 10% and 10% of an article are equivalent to a single price increase of
 (1) 19% (2) 20%
 (3) 21% (4) 22%
8. The price of an article was first increased by 10% and then again by 20%. If the last increased price be 33, the original price was
 (1) 30 (2) 27.50
 (3) 26.50 (4) 25
9. If a number is increased by 20% and the resulting number is again increased by 20%, what per cent is the total increase ?
 (1) 48% (2) 44%
 (3) 41% (4) 40%
10. A number is increased by 10% and then it is decreased by 10%. The net change in the number is
 (1) no increase or decrease
 (2) 2% decrease
 (3) 1% increase
 (4) 1% decrease
11. If radius of a circle is increased by 5%, then the increase in its area is
 (1) 10.25 % (2) 10 %
 (3) 5.75 % (4) 5 %
12. The percentage increase in the area of a rectangle, if each of its sides is increased by 20% is equal to
 (1) 32% (2) 34%
 (3) 42% (4) 44%
13. If the radius of a circle is decreased by 10%, then the area of the circle is decreased by
 (1) 89% (2) 18%
 (3) 19% (4) 25%
14. If the area of a square is increased by 44%, retaining its shape as a square, each of its sides increases by :
 (1) 19% (2) 21%
 (3) 22% (4) 20%
15. What will be the percentage increase in the area of a square when each of the its sides is increased by 10%?
 (1) 20 (2) 11
 (3) 121 (4) 21
16. If the radius of a cylinder is decreased by 50% and the height is increased by 50% to form a new cylinder, the volume will be decreased by
 (1) 0% (2) 25%
 (3) 62.5% (4) 75%
17. Each of the height and base-radius of a cone is increased by 100%. The percentage increase in the volume of the cone is
 (1) 700% (2) 400%
 (3) 300% (4) 100%
18. If both the radius and height of a right circular cone are increased by 20%, its volume will be increased by
 (1) 20% (2) 40%
 (3) 60% (4) 72.8%
19. If the height of a right circular cone is increased by 200% and the radius of the base is reduced by 50%, the volume of the cone
 (1) increases by 25% (2) increases by 50%
 (3) remains unaltered (4) decreases by 25%
20. If the height and the radius of the base of a cone are each increased by 100%, then the volume of the cone becomes
 (1) double that of the original
 (2) three times that of the original
 (3) six times that of the original
 (4) eight times that of the original
21. A sample of 50 litres of glycerin is found to be adulterated to the extent of 20%. How much pure glycerin should be added to it so as to bring down the percentage of impurity to 5% ?
 (1) 155 litres (2) 150 litres
 (3) 150.4 litres (4) 149 litres
22. 1 litre of water is added to 5 litres of alcohol-water solution containing 40% alcohol

strength. The strength of alcohol in the new solution will be

- (1) 30% (2) 33%
(3) 33.66% (4) 33.33%

23. If 4 litres of water is evaporated on boiling from 12 litres of salt solution containing 7 percentage salt, the percentage of salt in the remaining solution is

- (1) 10.5% (2) 11.5%
(3) 12% (4) 13%

24. A vessel has 60 litres of solution of acid and water having 80% acid. How much water be added to make it a solution in which acid forms 60%?

- (1) 48 litres (2) 20 litres
(3) 36 litres (4) None of these

25. 75 gm of sugar solution has 30% sugar in it. Then the quantity of sugar that should be added to the solution to make the quantity of the sugar 70% in the solution, is

- (1) 125 gm (2) 100 gm
(3) 120 gm (4) 130 gm

26. A litre of pure alcohol is added to 6 litres of 30% alcohol solution. The percentage of water in the solution is

- (1) 50% (2) 65%
(3) 60% (4) 40%

27. An ore contains 25% of an alloy that has 90% iron. Other than this, in the remaining 75% of the ore, there is no iron. To obtain 60 kg of pure iron, the quantity of the ore needed (in kgs) is approximately

- (1) 250.57 (2) 266.67
(3) 275.23 (4) 300

28. How much water must be added to 100 ml of 80 per cent solution of boric acid to reduce it to a 50 per cent solution?

- (1) 30 ml (2) 40 ml
(3) 50 ml (4) 60 ml

29. In one litre of a mixture of alcohol and water, water is 30%. The amount of alcohol that must be added to the mixture so that the part of water in the mixture becomes 15% is :

- (1) 1000 ml (2) 700 ml
(3) 300 ml (4) 900 ml

30. One type of liquid contains 20% water and the second type of liquid contains 35% of water. A glass is filled with 10 parts of first liquid and 4 parts of second liquid. The water in the new mixture in the glass is

- (1) 37% (2) 46%
(3) 12.14% (4) 24.28%

ANSWER KEY

1.	(1)	2.	(3)	3.	(3)	4.	(2)	5.	(1)	6.	(2)	7.	(3)	8.	(4)	9.	(2)	10.	(4)
11.	(1)	12.	(4)	13.	(3)	14.	(4)	15.	(4)	16.	(3)	17.	(1)	18.	(4)	19.	(4)	20.	(4)
21.	(2)	22.	(4)	23.	(1)	24.	(2)	25.	(2)	26.	(3)	27.	(2)	28.	(4)	29.	(1)	30.	(4)

PRACTICE SHEET 4

1. The present population of a city is 180000. If it increases at the rate of 10% per annum, its population after 2 years will be:

- (1) 207800 (2) 227800
(3) 217800 (4) 237800

2. The value of an equipment depreciates by 20% each year. How much less will the value of the equipment be after 3 years ?

- (1) 48.8% (2) 51.2%
(3) 54% (4) 60%

3. A district has 64000 inhabitants. If the population increases at the rate of 2.5% per annum, the number of inhabitants at the end of 3 years will be

- (1) 70000 (2) 69200
(3) 68921 (4) 68911

4. The value of a property depreciates every year by 10% of its value at the beginning of

the year. The present value of the property is 8100. What was its value 2 years ago?

- (1) 10,000 (2) 9000
(3) 9500 (4) 9801

5. The population of a town 2 years ago was 62,500. Due to migration to big cities, it decreases every year at the rate of 4%. The present population of the town is:

- (1) 57,600 (2) 56,700
(3) 76,000 (4) 75,000

6. The population of a town increases every year by 4%. If its present population is 50,000, then after 2 years it will be

- (1) 53,900 (2) 54,000
(3) 54,080 (4) 54,900

7. A man received 8,80,000 as his annual salary of the year 2007 which was 10% more than his annual salary in 2006. His annual salary in the year 2006 was

- (1) 4,80,000 (2) 8,00,000
(3) 4,00,000 (4) 8,40,000

8. Present population of a village is 67600. It has been increasing annually at the rate of 4%. What was the population of the village two years ago?

- (1) 62500 (2) 63000
(3) 64756 (4) 65200

9. The value of a machine depreciates by 5% every year. If its present value is 2,00,000, its value after 2 years will be

- (1) 1,80,500 (2) 1,99,000
(3) 1,80,000 (4) 2,10,000

10. The value of a property decreases every year at the rate of 5%. If its present value is 4,11,540, what was its value 3 years ago?

- (1) 4,50,000 (2) 4,60,000
(3) 4,75,000 (4) 4,80,000

11. 8% of the voters in an election did not cast their votes. In this election, there were only two candidates. The winner by obtaining 48% of the total votes defeated his contestant by 1100 votes. The total number of voters in the election was:

- (1) 21000 (2) 23500
(3) 22000 (4) 27500

12. In an election between two candidates, 75% of the voters cast their votes, out of which 2% votes were declared invalid. A candidate got 9261 votes which were 75% of the valid

votes. The total number of voters enrolled in that election was

- (1) 16000 (2) 16400
(3) 16800 (4) 18000

13. In an election between two candidates, the candidate getting 60% of the votes polled, is elected by a majority of 14,000 votes. The number of votes polled by the winning candidate is

- (1) 28,000 (2) 32,000
(3) 42,000 (4) 46,000

14. In an office 40% of the staff is female, 40% of the females and 60% of the males voted for me. The percentage of votes I got was

- (1) 24% (2) 42%
(3) 50% (4) 52%

15. In an election there were only two candidates. One of the candidates secured 40% of votes and is defeated by the other candidate by 298 votes. The total number of votes polled is

- (1) 745 (2) 1460
(3) 1490 (4) 1500

16. In an assembly election, a candidate got 55% of the total valid votes. 2% of the total votes were declared invalid. If the total number of voters is 104000, then the number of valid votes polled in favour of the candidate is:

- (1) 56506 (2) 56650
(3) 56560 (4) 56056

17. Two candidates contested in an election. One got 60% of the votes and won by 1600 votes. What is the number of votes polled?

- (1) 9000 (2) 8000
(3) 10000 (4) 7500

18. In an election, three candidates contested. The first candidate got 40% votes and the second got 36% votes. If total number of votes polled were 36000, find the number of votes got by the 3rd candidate.

- (1) 8040 (2) 8640
(3) 9360 (4) 9640

19. Two persons contested an election of Parliament. The winning candidate secured 57% of the total votes polled and won by a majority of 42,000 votes. The number of total votes polled is

- (1) 5,00,000 (2) 6,00,000
(3) 3,00,000 (4) 4,00,000

- 20.** In an election, a candidate who gets 84 % of the votes is elected by a majority of 476 votes. What is the total number of votes polled?
 (1) 900 (2) 810
 (3) 600 (4) 700
- 21.** In an examination 34% failed in Mathematics and 42% failed in English. If 20% failed in both the subjects, the percentage of students who passed in both subjects was
 (1) 54% (2) 50%
 (3) 44% (4) 56%
- 22.** A candidate secured 30% marks in an examination and failed by 6 marks. Another secured 40% marks and got 6 marks more than the bare minimum to pass. The maximum marks are
 (1) 150 (2) 120
 (3) 100 (4) 180
- 23.** In an examination, 52% students failed in Hindi and 42% in English. If 17% failed in both the subjects, what percentage of students passed in both the subjects?
 (1) 38% (2) 33%
 (3) 23% (4) 18%
- 24.** In a group of students, 70% can speak English and 65% can speak Hindi. If 27% of the students can speak none of the two languages, then what per cent of the group can speak both the languages?
 (1) 38% (2) 62%
 (3) 28% (4) 23%
- 25.** 25% of the candidates who appeared in an examination failed to qualify and only 450 candidates qualified. The number of candidates, who appeared in the examination, was
 (1) 700 (2) 600
 (3) 550 (4) 500
- 26.** In two successive years 100 and 75 students of a school appeared at the final examination. Respectively, 75% and 60% of them passed. The average rate of pass is
 (1) 68.57 % (2) 78%
 (3) 80.5 % (4) 80%
- 27.** A student has to secure minimum 35% marks to pass in an examination. If he gets 200 marks and fails by 10 marks, then the maximum marks are
 (1) 300 (2) 400
 (3) 500 (4) 600
- 28.** A candidate who scores 30 per cent fails by 5 marks, while another candidate who scores 40 per cent marks gets 10 more than minimum pass marks. The minimum marks required to pass are
 (1) 50 (2) 70
 (3) 100 (4) 150
- 29.** In an examination, 60% of the candidates passed in English and 70% of the candidates passed in Mathematics, but 20% failed in both of these subjects. If 2500 candidates passed in both the subjects, the number of candidates who appeared at the examination was
 (1) 3000 (2) 3500
 (3) 4000 (4) 5000
- 30.** In a test a student got 30% marks and failed by 25 marks. In the same test another student got 40% marks and secured 25 marks more than the essential minimum pass marks. The maximum marks for the test were
 (1) 400 (2) 480
 (3) 500 (4) 580

ANSWER KEY

1.	(3)	2.	(1)	3.	(3)	4.	(1)	5.	(1)	6.	(3)	7.	(2)	8.	(1)	9.	(1)	10.	(4)
11.	(4)	12.	(3)	13.	(3)	14.	(4)	15.	(3)	16.	(4)	17.	(2)	18.	(2)	19.	(3)	20.	(4)
21.	(3)	22.	(2)	23.	(3)	24.	(2)	25.	(2)	26.	(1)	27.	(4)	28.	(1)	29.	(4)	30.	(3)

PRACTICE SHEET 5

1. Price of sugar rises by 20%. By how much percent should the consumption of sugar be reduced so that the expenditure does not change?
 (1) 20% (2) 10%
 (3) $16\frac{2}{3}\%$ (4) 15%
2. If food prices go up by 10%, by how much should a man reduce his consumption so as not to increase his expenditure?
 (1) $9\frac{1}{11}\%$
 (2) 10%
 (3) $11\frac{1}{9}\%$
 (4) The data is not sufficient
3. In the new budget, the price of kerosene oil rose by 25%. By how much per cent must a person reduce his consumption of kerosene oil so that his expenditure on it does not increase?
 (1) 20% (2) 25%
 (3) 50% (4) 40%
4. If the price of tea is increased by 20%, by how much per cent the consumption of tea be reduced so that there is no increase in the expenditure on it?
 (1) $83\frac{1}{3}\%$ (2) 20%
 (3) $16\frac{2}{3}\%$ (4) $8\frac{1}{3}\%$
5. If the price of a commodity is decreased by 20% and its consumption is increased by 20%, what will be the increase or decrease in the expenditure on the commodity?
 (1) 4% increase (2) 4% decrease
 (3) 8% decrease (4) 8% increase
6. The price of a certain item is increased by 15%. If a consumer wants to keep his expenditure on the item the same as before, how much per cent must he reduce his consumption of that item?
 (1) 15% (2) $13\frac{1}{23}\%$
- (3) $16\frac{2}{3}\%$ (4) $10\frac{20}{23}\%$
7. If the price of a commodity is increased by 50%, by what fraction must its consumption be reduced so as to keep the same expenditure on its consumption?
 (1) $\frac{1}{4}$ (2) $\frac{1}{3}$
 (3) $\frac{1}{2}$ (4) $\frac{2}{3}$
8. If the price of rice be raised by 25%, the percent by which a house-holder must reduce his consumption of rice so as not to increase his expenditure on rice is
 (1) 225% (2) 25.75%
 (3) 25% (4) 20%
9. If the duty on an article is reduced by 40% of its present rate, by how much per cent must its consumption increase in order that the revenue remains unaltered
 (1) 60% (2) $62\frac{1}{3}\%$
 (3) 72% (4) $66\frac{2}{3}\%$
10. Price of milk has increased by 20%. To keep the expenditure unchanged, the present consumption s to be reduced by :
 (1) 20% (2) 18%
 (3) 10% (4) $16\frac{2}{3}\%$
11. The Government reduced the price of sugar by 10 per cent. By this a consumer can buy 6.2 kg more sugar for 837. The reduced price per kg of sugar is
 (1) 12.50 (2) 13.00
 (3) 13.50 (4) 14.00
12. The price of sugar is reduced by 20%. Now a person can buy 500g more sugar for 36. The original price of the sugar per kilogram was
 (1) 14.40 (2) 18
 (3) 15.60 (4) 16.50
13. A reduction of 10% in the price of sugar enables a housewife to buy 6.2 kg more for 1116. The reduced price per kg is
 (1) 12 (2) 14
 (3) 16 (4) 18

- 14.** When the price of sugar decreases by 10%, a man could buy 1 kg more for 270. Then the original price of sugar per kg is
 (1) 25 (2) 30
 (3) 27 (4) 32
- 15.** A reduction of 20% in the price of an apple enables a man to buy 10 apples more for 54. The reduced price of apples per dozen is
 (1) 4.32 (2) 12.96
 (3) 10.80 (4) 14.40
- 16.** Due to an increase of 50% in the price of eggs, 4 eggs less are available for 24. The present rate of eggs per dozen is :
 (1) 24 (2) 27
 (3) 36 (4) 42
- 17.** A reduction of 20% in the price of wheat enables Lalita to buy 5 kg more wheat for ₹ 320. The original rate (in rupees per kg) of wheat was
 (1) 16 (2) 18
 (3) 20 (4) 21
- 18.** Due to an increase of 20% in the price of eggs, 2 eggs less are available for 24. The present rate of eggs per dozen is :
 (1) 25.00 (2) 26.20
 (3) 27.80 (4) 28.80
- 19.** A reduction of 25% in the price of rice enables a person to buy 10 kg more rice for ₹ 600. The reduced per kg price of rice is
 (1) ₹ 30 (2) ₹ 25
 (3) ₹ 20 (4) ₹ 15
- 20.** A reduction in the price of apples enables a person to purchase 3 apples for ₹ 1 instead of ₹ 1.25. What is the % of reduction in price (approximately)?
 (1) 20% (2) 25%
 (3) 30% (4) $33\frac{1}{3}\%$
- 21.** A number, on subtracting 15 from it, reduces to its 80%. What is 40% of the number?
 (1) 75 (2) 60
 (3) 30 (4) 90
- 22.** A reduction of 21% in the price of an item enables a person to buy 3 kg more for ₹ 100. The reduced price of item per kg is
 (1) 5.50 (2) 7.50
 (3) 10.50 (4) 7.00
- 23.** A reduction of 20% in the price of sugar enables a purchaser to obtain 8 kg more for ₹. 160. Then the price per kg before reduction was
 (1) ₹. 5 (2) ₹. 6
 (3) ₹. 10 (4) ₹. 4
- 24.** The price of an item was increased by 10%. This reduced the monthly total sales by 20%. The overall effect on the value of monthly sales is a
 (1) 10% increase (2) 10% decrease
 (3) 12% increase (4) 12% decrease
- 25.** The price of rice has increased by 60%. In order to restore the original price, the new price must be reduced by
 (1) $33\frac{1}{3}\%$ (2) $37\frac{1}{2}\%$
 (3) 40% (4) 45%
- 26.** In a village, each of the 60% of families has a cow; each of the 30% of families has a buffalo and each of the 15% of families has both a cow and buffalo. In all there are 96 families in the village. How many families do not have a cow or a buffalo?
 (1) 20 (2) 24
 (3) 26 (4) 28
- 27.** A man invests a part of ₹ 10,000 at 5% and the remainder at 6%. The 5% investment yields annually ₹ 76.50 more than the 6% investment. The amount invested at 6% is
 (1) 3,600 (2) 3,550
 (3) 3,850 (4) 4,000
- 28.** A number is divided into two parts in such a way that 80% of 1st part is 3 more than 60% of 2nd part and 80% of 2nd part is 6 more than 90% of the 1st part. Then the number is
 (1) 125 (2) 130
 (3) 135 (4) 145
- 29.** Price of a one gram gold coin decreased by 10% on its initial price on Monday and increased by 20% on Tuesday and again increased by 8% on Wednesday, and 5% increase on Thursday. If the final price on Thursday is ₹ 5511.24, then the initial price (in ₹) of one gram gold coin on Monday was?
 (a) 4500 (b) 4250
 (c) 4000 (d) 5000

30. Radha saves $x\%$ of her income. If her income increases by 28% and the expenditure increases by 20%, then her savings increases by 40%. What is the value of x ?

- (a) 35
(b) 40
(c) 50
(d) 25

ANSWER KEY

1.	(3)	2.	(1)	3.	(1)	4.	(3)	5.	(2)	6.	(2)	7.	(2)	8.	(4)	9.	(4)	10.	(4)
11.	(3)	12.	(2)	13.	(4)	14.	(2)	15.	(2)	16.	(3)	17.	(1)	18.	(4)	19.	(4)	20.	(1)
21.	(3)	22.	(4)	23.	(1)	24.	(4)	25.	(2)	26.	(2)	27.	(3)	28.	(3)	29.	(1)	30.	(2)

CDS PYQ

1. A person's salary has increased from Rs 7200 to Rs 8100. What is percentage increase in his salary?

[CDS 2013 (II)]

- (a) 25%
(b) 18%
(c) $16\frac{2}{3}\%$
(d) 12.5%

2. A water pipe is cut into two pieces. The longer piece is 70% of the length of the pipe. By how much percentage is the longer piece longer than the shorter piece.

[CDS 2014 (I)]

- (a) 140%
(b) $400/3\%$
(c) 40%
(d) None of these

3. If $m\%$ of $m + n\%$ of $n = 2\%$ of mn , then what percent of m is n

[CDS 2014 (II)]

- (a) 50%
(b) 75%
(c) 100%
(d) Cannot determined

4. If side of a cube is increased by 100% , then by what percentage is the surface area of the cube increased ?

[CDS 2014 (II)]

- (a) 150%
(b) 200%
(c) 300%
(d) 400%

5. The price of commodity increased by 5% from 2010 to 2011, 8% from 2011 to 2012 and 7% from 2012 to 2013. What is the average price increase(approx.) from 2010 to 2013?

[CDS 2014 (II)]

- (a) 26%
(b) 32%
(c) 24%
(d) 30%

6. The radius of a circle is increased by 6%, then the area will increase by

[CDS 2014 (II)]

- (a) 6%
(b) 9%
(c) 12.36%
(d) 16.64%

7. A person could save 10% of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased?

[CDS 2015 (I)]

- (a) $22\frac{2}{9}\%$
(b) $23\frac{1}{3}\%$
(c) $24\frac{2}{9}\%$
(d) $25\frac{2}{9}\%$

8. 20% of a number when added to 20 becomes the number itself, then the number is

[CDS 2015 (II)]

- (a) 20
(b) 25
(c) 50
(d) 80

9. A's salary was increased by 40% and then decreased by 20%. On the whole A's salary is increased by

[CDS 2015 (II)]

- (a) 60%
(b) 40%
(c) 20%
(d) 12%

10. In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. There were only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 voters more than his rival. The number of voters on the voter list is

[CDS 2015 (II)]

- (a) 3600 (b) 6200
(c) 6028 (d) 6400

11. The salary of a person is increased by 10% of his original salary. But he received the same amount even after increment. What is the percentage of his salary he did not receive?

[CDS 2016 (I)]

- (a) 11% (b) 10%
(c) 100/11% (d) 90/11%

12. The expenditure of a household for a certain month is Rs 20,000 out of which Rs 8000 is spent on education, Rs 5900 on food, Rs 2800 on shopping and the rest on personal care. What percentage of expenditure is spent on personal care?

[CDS 2016 (I)]

- (a) 12% (b) 16.5%
(c) 18% (d) 21.8%

13. A candidate scoring $x\%$ marks in an examination fails by a marks, while another candidate who scores $y\%$ marks gets b marks more than the minimum required pass marks. What is the maximum marks for the examination?

[CDS 2016 (II)]

- (a) $\frac{100(a+b)}{x-y}$ (b) $\frac{100(a-b)}{x+y}$
(c) $\frac{100(a+b)}{y-x}$ (d) $\frac{100(a-b)}{x-y}$

14. If $a\%$ of $a + b\%$ of $b = 2\%$ of ab , then what percent of a is b ?

[CDS 2017 (I)]

- (a) 50%
(b) 75%
(c) 100%
(d) Cannot be determined

15. $\frac{5}{9}$ th part of the population in a village are males. If 30% of the males are married, the percentage of unmarried females in the total population is?

[CDS 2017 (I)]

- (a) $20\frac{2}{9}\%$ (b) $27\frac{2}{9}\%$
(c) $27\frac{7}{9}\%$ (d) $29\frac{2}{9}\%$

16. A fruit seller has a certain number of mangoes of which 5% are rotten. He sells 75% of the remainder and he is left with 95 mangoes. How many mangoes did he have originally?

[CDS 2017 (II)]

- (a) 500 (b) 450
(c) 400 (d) 350

17. A student has to secure 40% of marks to pass an examination. He gets only 45 marks and fails by 5 marks. The maximum marks are

[CDS 2017 (II)]

- (a) 120 (b) 125
(c) 130 (d) 150

18. When prices rise by 12%, if the expenditure is to be the same, what is the percentage of consumption to be reduced?

[CDS 2017 (II)]

- (a) $16\frac{2}{3}\%$ (b) $10\frac{2}{7}\%$
(c) $16\frac{3}{5}\%$ (d) $10\frac{5}{7}\%$

19. If the price of wheat rises by 25%, then by how much percent must a man reduce his consumption in order to keep his budget the same as before?

[CDS 2018 (I)]

- (a) 15% (b) 20%
(c) 25% (d) 30%

20. A person bought two articles X and Y from a departmental store. The sum of prices before sales tax was Rs 130. There was no sales tax on the article X and 9% sales tax on the article Y. The total amount the person paid, including the sales tax was Rs 136.75. What was the price of the article Y before sales tax?

[CDS 2018 (II)]

- (a) Rs 75 (b) Rs 85
(c) Rs 122 (d) Rs 125

21. In the hostel the rent per room is increased by 20%. If the number of rooms in the hostel is also increased by 20% and the hostel is always full, then what is the percentage change in the total collection at the cash counter?

[CDS 2019 (I)]

- (a) 30% (b) 40%
(c) 44% (d) 48%

- 22.** x, y and z are three numbers such that x is 30% of z and y is 40% of z. If x is p% of y, then what is the value of p?
[CDS 2019 (I)]
(a) 45 (b) 55
(c) 65 (d) 75
- 23.** The train fare and bus fare between two stations is in the ratio 3 : 4, If the train fare increases by 20% and bus fare increases by 30% , then what is the ratio between revised train fare and revised bus fare ?
[CDS 2019 (II)]
(a) 9/13 (b) 17/12
(c) 32/43 (d) 19/21
- 24.** If the annual income of X is 20% more than that of Y, then the income of Y is less than that of X by p%. What is the value of p?
[CDS 2020 (I)]
(a) 10 (b) $16\frac{2}{3}$
(c) $17\frac{1}{3}$ (d) 20
- 25.** The price of an article X is increases by 20% every year and price of article Y increases by 10% every year. In the year 2010, the price of article X was Rs 5000 and price of article Y was Rs 2000. In which year the difference in their prices exceeded Rs 5000 for the first time?
[CDS 2020 (I)]
(a) 2012 (b) 2013
(c) 2014 (d) 2015
- 26.** The number of items in a booklet is N. In the first year there is an increase of x% in this number and in the subsequent year there is a decrease of x%. At the end of the two years, what will be the number of items in the booklet?
[CDS 2020 (II)]
(a) Less than N (b) Equal to N
(c) More than N (d) It depends on N
- 27.** X sales his goods 25% cheaper than Y and 25% dearer than Z, How much percentage is Z's goods cheaper than Y?
[CDS 2021 (I)]
(a) 100/3% (b) 40%
(c) 50% (d) 200/3%
- 28.** In a mixture of 80 litres of a liquid and water, 25% of the mixture is the liquid. How much water should be added to the mixture so that the liquid becomes 20% of the mixture?
[CDS 2021 (I)]
(a) 15 litres (b) 20 litres
(c) 24 litres (d) 25 litres
- 29.** If p varies directly as q and inversely as square of r, what is the percentage increase in p due to an increase in q by 20% and a decrease in r by 20%?
[CDS 2021 (II)]
(a) 87.5% (b) 85%
(c) 82.5% (d) 80%

ANSWER KEY

1.	(d)	2.	(b)	3.	(c)	4.	(c)	5.	(a)	6.	(c)	7.	(a)	8.	(b)	9.	(d)	10.	(b)
11.	(c)	12.	(b)	13.	(c)	14.	(c)	15.	(c)	16.	(c)	17.	(b)	18.	(d)	19.	(b)	20.	(a)
21.	(c)	22.	(d)	23.	(a)	24.	(b)	25.	(b)	26.	(a)	27.	(b)	28.	(b)	29.	(a)		

CAPF PYQ

1. How much water to be added to 75ml of alcohol so that the mixture contains 25% of alcohol?

[CAPF 2021]

- (a) 100ml (b) 225 ml
(c) 250 ml (d) 125 ml

2. If 35% of a number is 416 more than 27% of the same number, then the number is

[CAPF 2021]

- (a) 5200 (b) 2600
(c) 3900 (d) 3328

3. A tree is at present 9 feet tall, every year it grows $\frac{1}{9}$ of its height, will be the height of the tree three years?

[CAPF 2021]

- (a) 12 feet (b) 12.34 feet
(c) 13 feet (d) 13.10 feet

ANSWER KEY

1.	(b)	2.	(a)	3.	(b)														
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